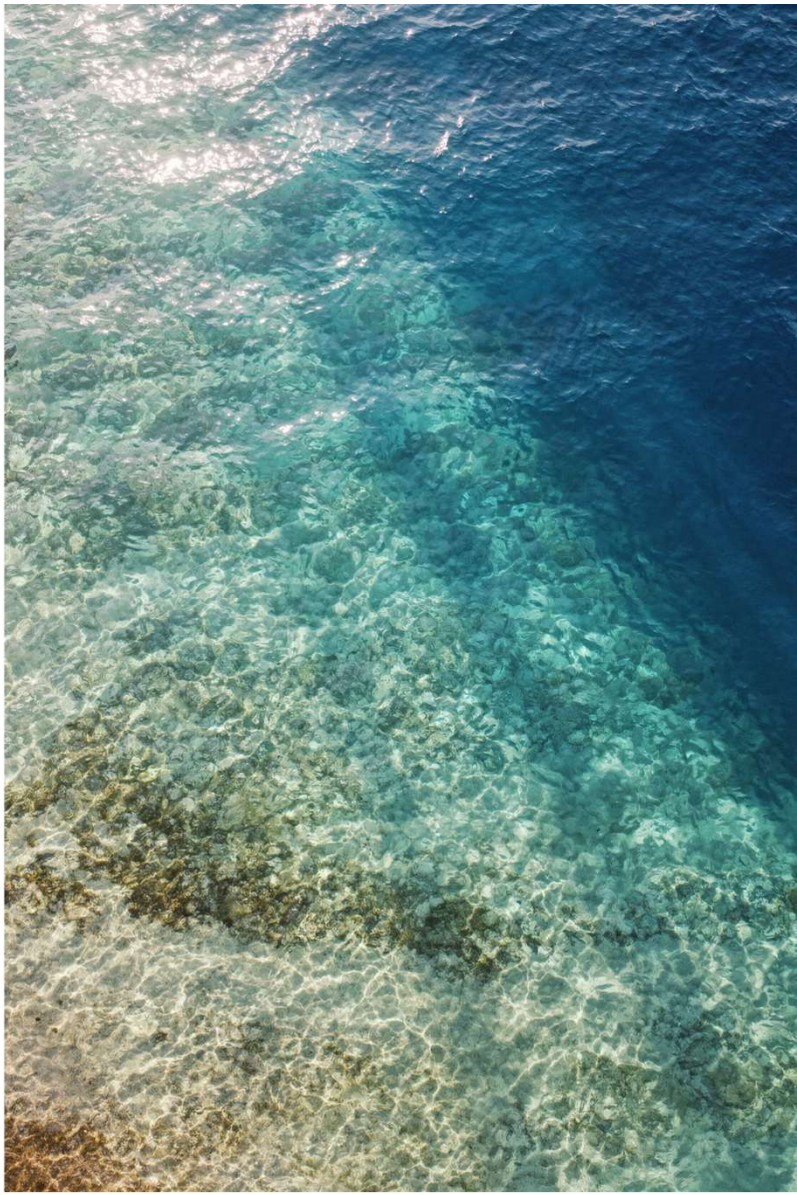
Ocean Thermal Energy Conversion (OTEC): Harnessing the Ocean's Thermal Gradient for Sustainable Power



CHAPTER 1

The Promise of Ocean Thermal Energy

The world's oceans hold an extraordinary reservoir of renewable energy, waiting to be unlocked through innovative technology that transforms temperature differences into clean electricity.

The Ocean's Untapped Power

Thermal Storage

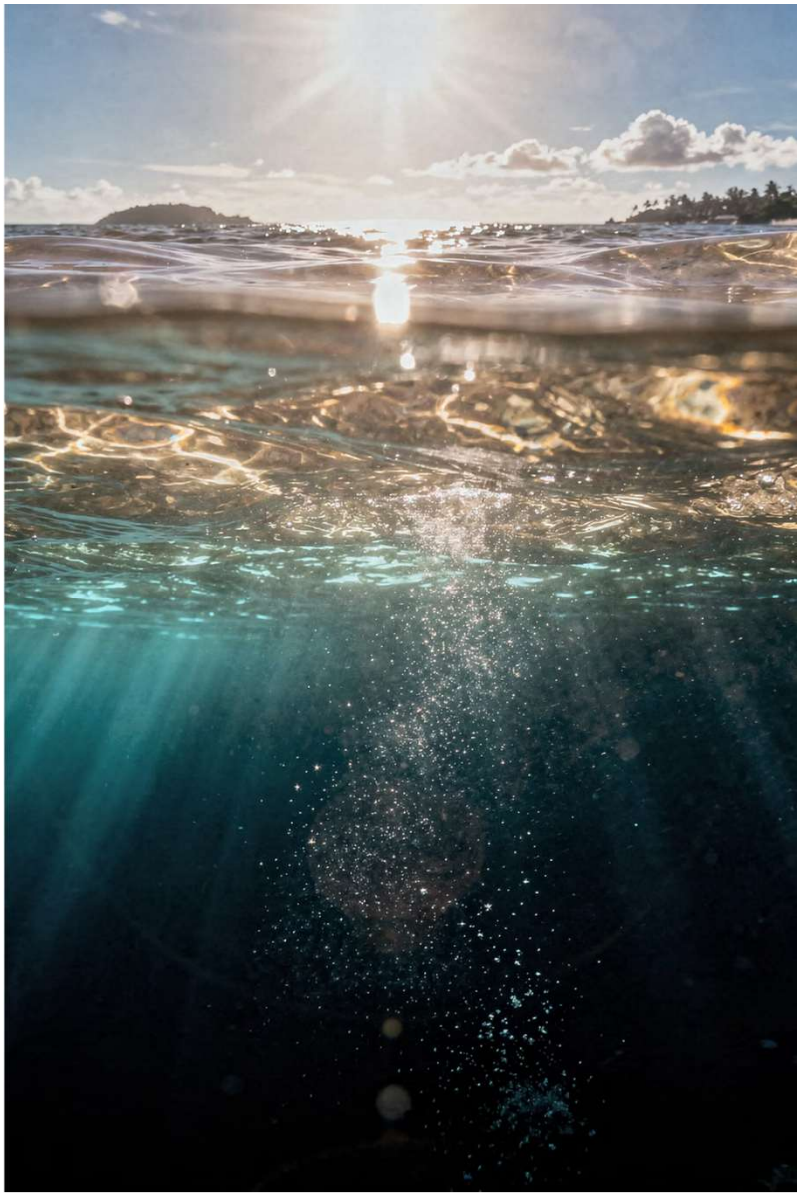
Surface waters maintain temperatures of 20–30°C whilst deep waters remain at a frigid 4–6°C, creating a persistent thermal gradient.

Global Potential

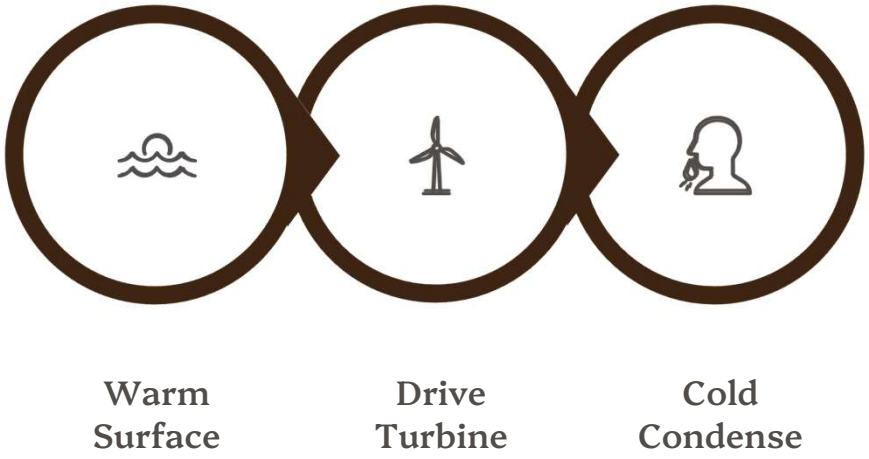
Theoretical OTEC capacity reaches 8–10 terawatts globally—sufficient to power millions of homes with clean, renewable energy.

Tropical Advantage

Oceans between 25°N and 25°S latitudes offer the most stable and exploitable temperature differences for continuous power generation.



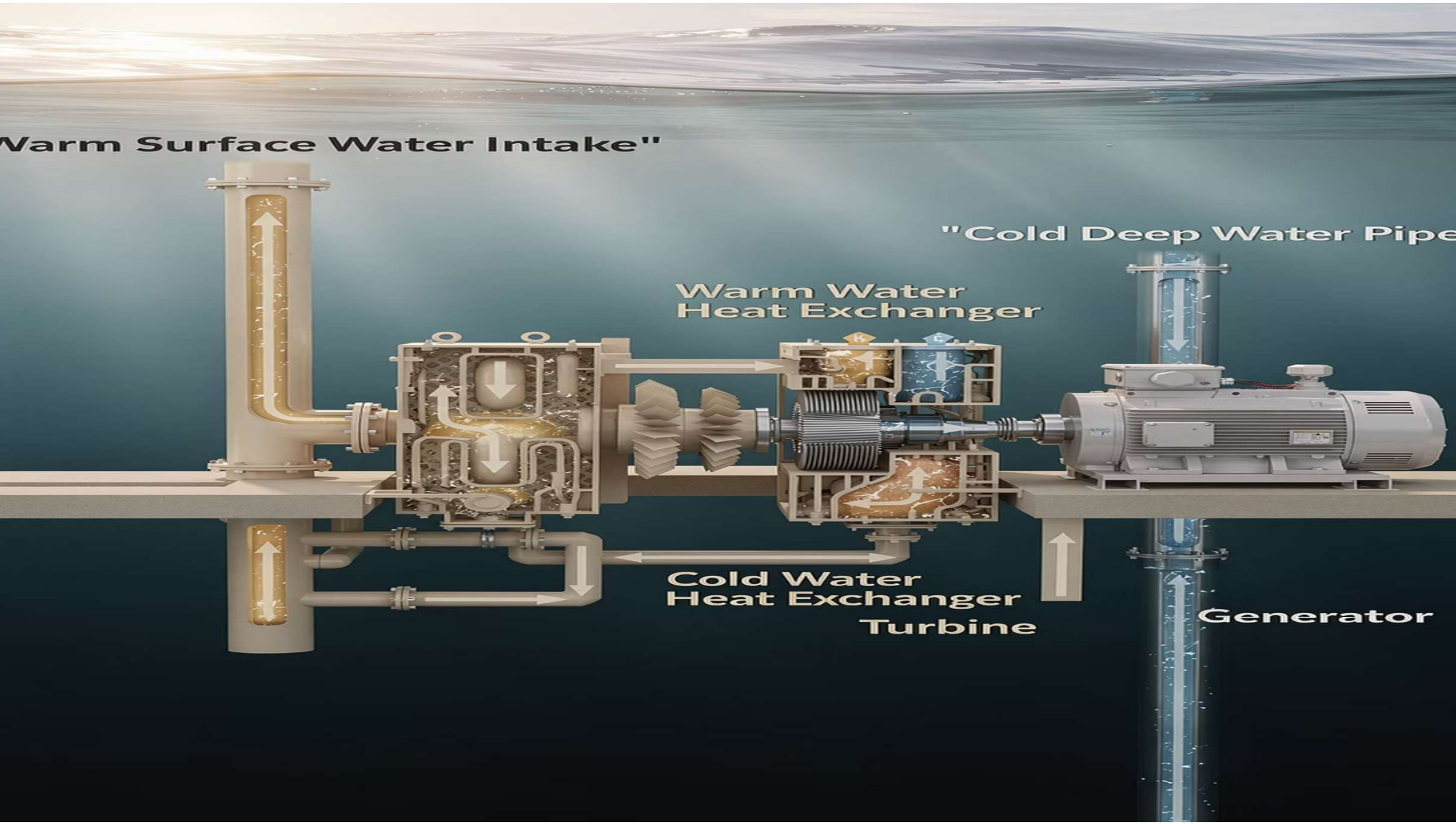
How OTEC Works: The Principle of Operation



The continuous cycle operates 24 hours a day, providing reliable baseload power unlike intermittent solar or wind sources.

OTEC technology exploits the temperature difference between warm surface seawater and cold deep seawater to drive a sophisticated heat engine. This thermal gradient converts heat energy into mechanical energy, which then generates electricity.

Whilst efficiency remains modest at approximately 6–8% due to the relatively small temperature differences involved, the energy source is entirely renewable and remarkably constant throughout the year.

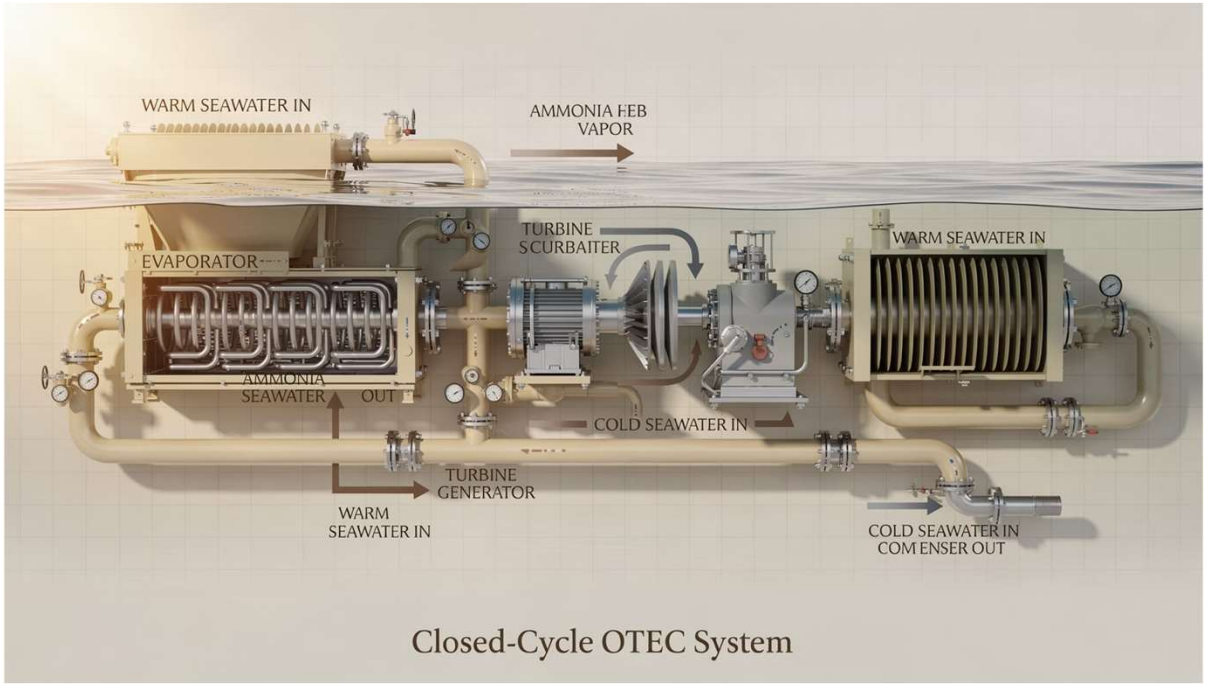

Warm Surface Water Intake"
"Cold Deep Water Pipe
Warm Water
Heat Exchanger
Cold Water
Heat Exchanger
Turbine
Generator

CHAPTER 2

Classification of OTEC Systems

Three distinct technological approaches have emerged, each with unique advantages for converting oceanic thermal gradients into usable energy and valuable by-products.

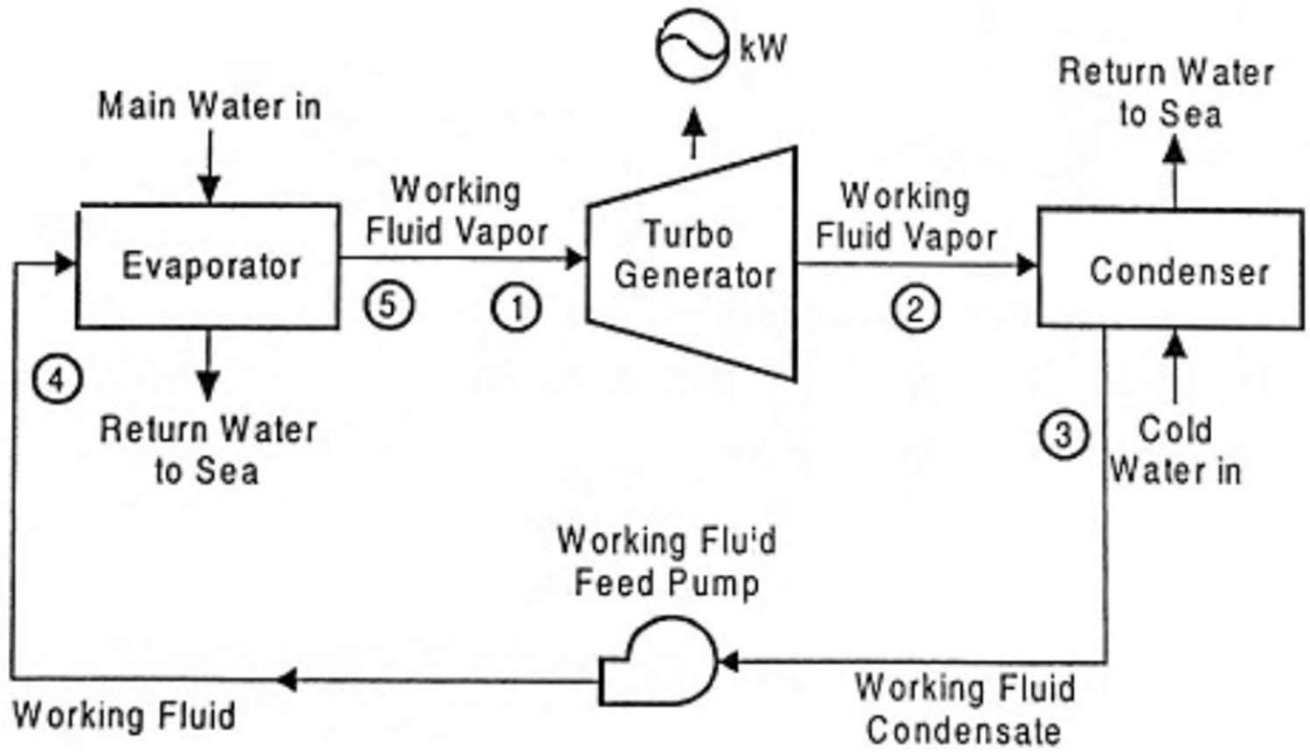
Closed-Cycle OTEC



Closed-Cycle OTEC System

The closed-cycle system employs a working fluid with an extremely low boiling point, typically ammonia or similar refrigerants. Warm surface seawater vaporises this fluid within a heat exchanger, and the resulting vapour pressure drives a turbine to generate electricity.

Cold deep seawater then condenses the vapour back into liquid form, allowing continuous reuse within the sealed system. This approach represents the most extensively researched and widely implemented OTEC configuration worldwide.


kW
Main Water in
Working
Fluid Vapor
Turbo
Generator
Working
Fluid Vapor
Return Water
to Sea
Evaporator
Condenser
5
1
2
4
3
Return Water
to Sea
Cold
Water in
Working Fluid
Feed Pump
Working Fluid
Working Fluid
Condensate

The closed-cycle system uses a working fluid, like ammonia, pumped around a closed loop, which has three components: a pump, turbine and heat exchangers (evaporator and condenser).

Warm seawater passing through the evaporator converting the ammonia liquid (4) into high-pressure ammonia vapour at (5).

The high-pressure vapour at (1) is then fed into an expander where it passes through and rotates a turbine connected to a generator.

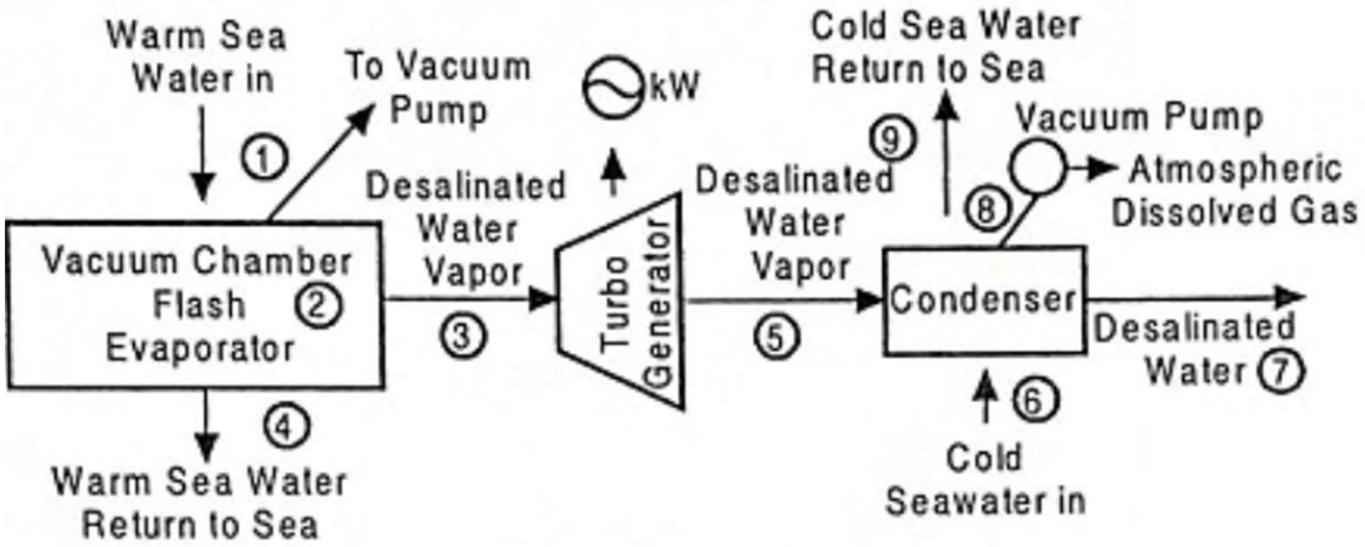
Low-pressure ammonia vapour leaving the turbine (2) is passed through a condenser, where the cold seawater cools the ammonia, returning the ammonia back into a liquid (3).

Open-Cycle OTEC

In the open-cycle approach, warm surface seawater enters a low-pressure vacuum chamber where it spontaneously boils, producing steam directly from seawater. This steam expands through a turbine, generating mechanical power for electricity production.

The steam subsequently condenses when exposed to cold deep seawater, yielding a valuable secondary product: fresh, desalinated water suitable for drinking or irrigation—a critical benefit for water-scarce tropical islands.




Warm Sea
Water in
①
To Vacuum
Pump
kW
Cold Sea Water
Return to Sea
⑨
Vacuum Pump
Atmospheric
Dissolved Gas
⑧
Desalinated
Water
Vapor
③
Turbo
Generator
Desalinated
Water
Vapor
⑤
Vacuum Chamber
Flash
Evaporator
②
Condenser
Desalinated
Water ⑦
④
Warm Sea Water
Return to Sea
⑥
Cold
Seawater in

The open-cycle system uses the warm seawater as the working fluid.

The warm seawater passing through the evaporator (2) is converted into steam (3), which drives the turbine/generator.

Pressure differential is maintained across the turbine due to the lower temperature on the condenser side

After leaving the turbine (5), the steam is cooled by the cold seawater to form desalinated water.

The desalinated water is pure fresh water for domestic and commercial use.

Hybrid-Cycle OTEC

Dual Process

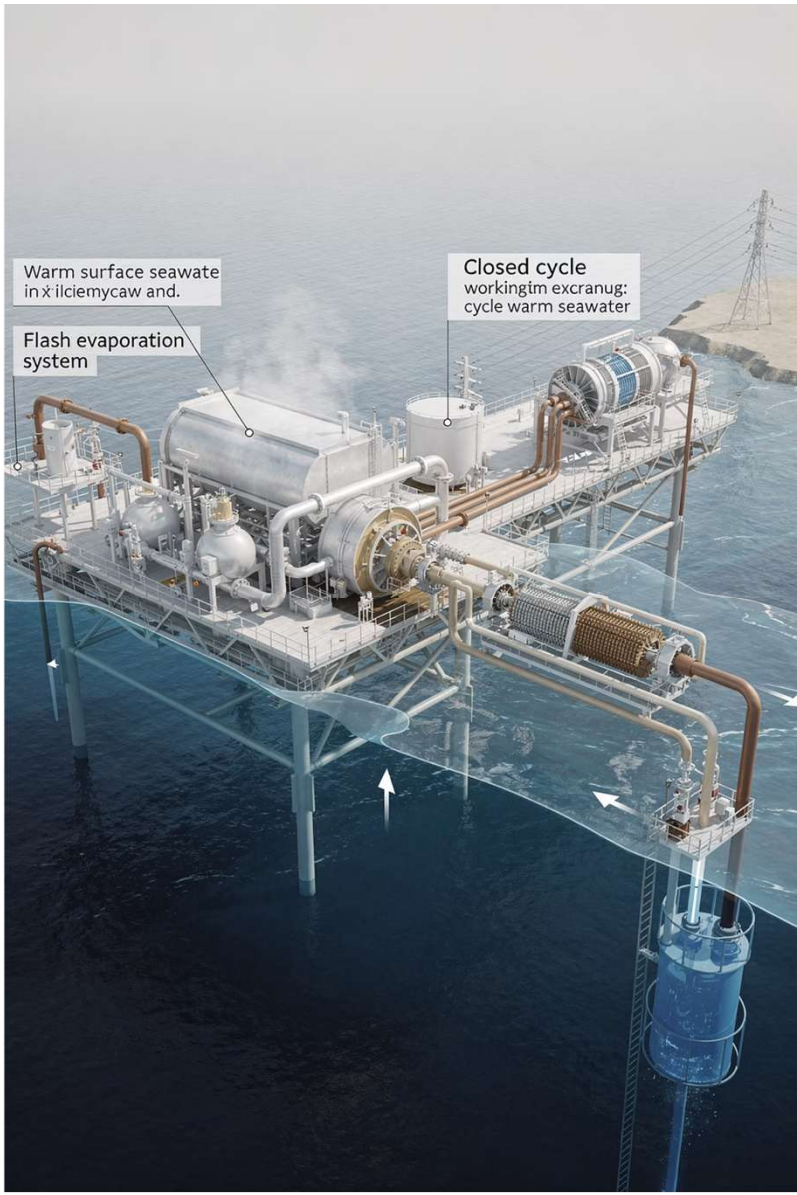
Combines the best features of both closed and open cycles for optimised performance and versatility.

Operation Method

Warm seawater flash-evaporates to steam, which then vaporises a secondary working fluid that drives the turbine generator.

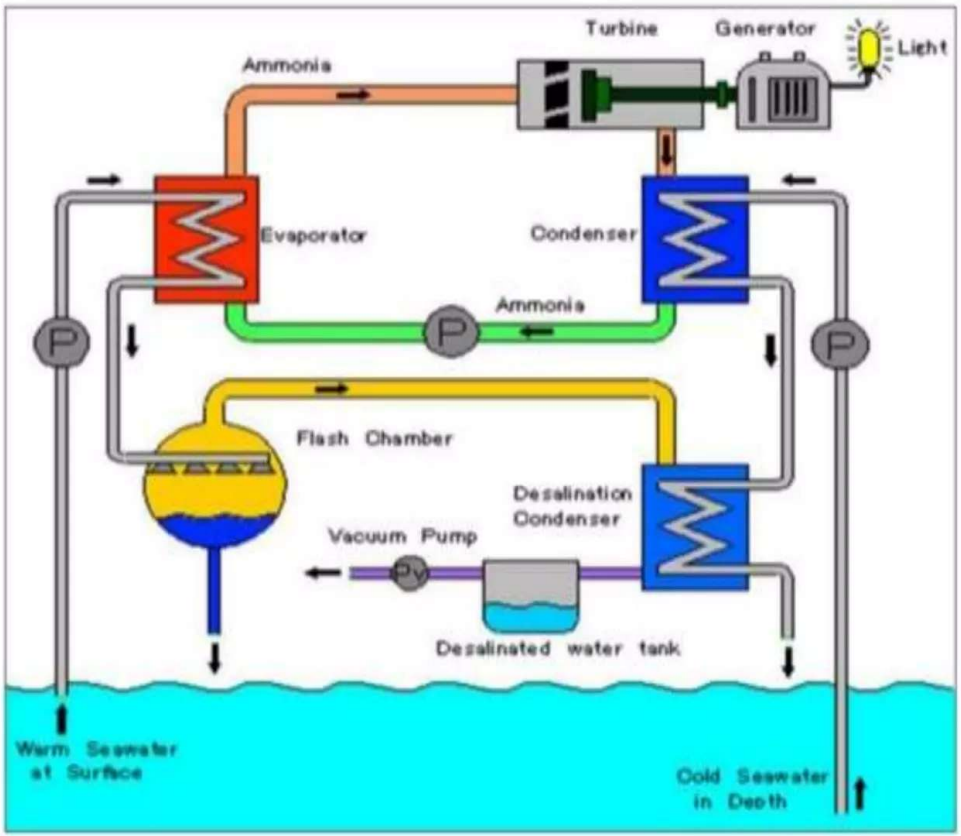
Economic Advantage

Produces both electricity and desalinated fresh water simultaneously, significantly enhancing overall economic viability and community value.



3) Hybrid Cycle OTEC

- Combination of Closed & Open Cycle



The hybrid system uses parts of both open-cycle and closed-cycle systems to produce electricity and desalinated water.

In this arrangement, electricity is generated in the closed-cycle system and

the warm and cold seawater discharges are passed through the flash evaporator and condenser of the open-cycle system to produce fresh water.

Advantages of OTEC Power Plant:

- It is clean form of energy conversion.
- It does not occupy land areas.
- No payment for the energy required.
- It can be a steady source of energy since the temperature are almost steady.

Disadvantages:

- About 30% of power produced will be used to pump water.
- The system should to withstand strong harmful effects of sea water. (hurricane, debris, fishes)
- The material used should withstand corrosive atmosphere.
- Very high investment is required.
- Plant size is limited to 100 MW due to large size of components.

CHAPTER 3

OTEC Power Stations Around the World

From pioneering experiments in the 1930s to cutting-edge floating platforms today, OTEC technology has evolved through decades of innovation and international collaboration.

Historic Milestones: The Pioneers

1930

Georges Claude constructed the world's first OTEC plant in Matanzas, Cuba, successfully generating 22 kilowatts of electricity and proving the fundamental concept.

1

2

1979

Lockheed Martin's Mini-OTEC project produced 18 kilowatts of net power off the coast of Hawaii, establishing the technical feasibility of floating offshore OTEC platforms.

Modern Operational Plants



Japan's Saga University

Maintains an ongoing pilot plant conducting cutting-edge research and demonstrations of closed-cycle OTEC technology with advanced materials.



Makai Ocean Engineering

Hawaii-based facility rigorously testing next-generation heat exchangers and innovative cold water pipe designs for enhanced efficiency.



K-OTEC1000 Platform

South Korea's groundbreaking 2019 deployment represents the world's first commercial-scale floating OTEC platform, marking a new era.

Emerging Projects and Global Interest

Growing Global Movement

Nations worldwide recognise OTEC's potential for energy independence and climate resilience, driving substantial investment in research and development.

- ***India's National Institute of Ocean Technology (NIOT) developing a 1 megawatt OTEC plant off the Tamil Nadu coast***
- ***Caribbean and Pacific territories including Hawaii, Puerto Rico, and Guam actively exploring OTEC for energy sovereignty***
- ***China and France investing in innovative hybrid OTEC systems integrating sustainable aquaculture and desalination capabilities***


K-OTEC1000

CHAPTER 4

Environmental Impacts and Considerations

Like all energy technologies, OTEC presents both environmental opportunities and challenges that require careful assessment, thoughtful design, and responsible management.

Potential Environmental Benefits



Zero Emissions

Entirely renewable with no greenhouse gas emissions during operation, contributing directly to climate change mitigation efforts.



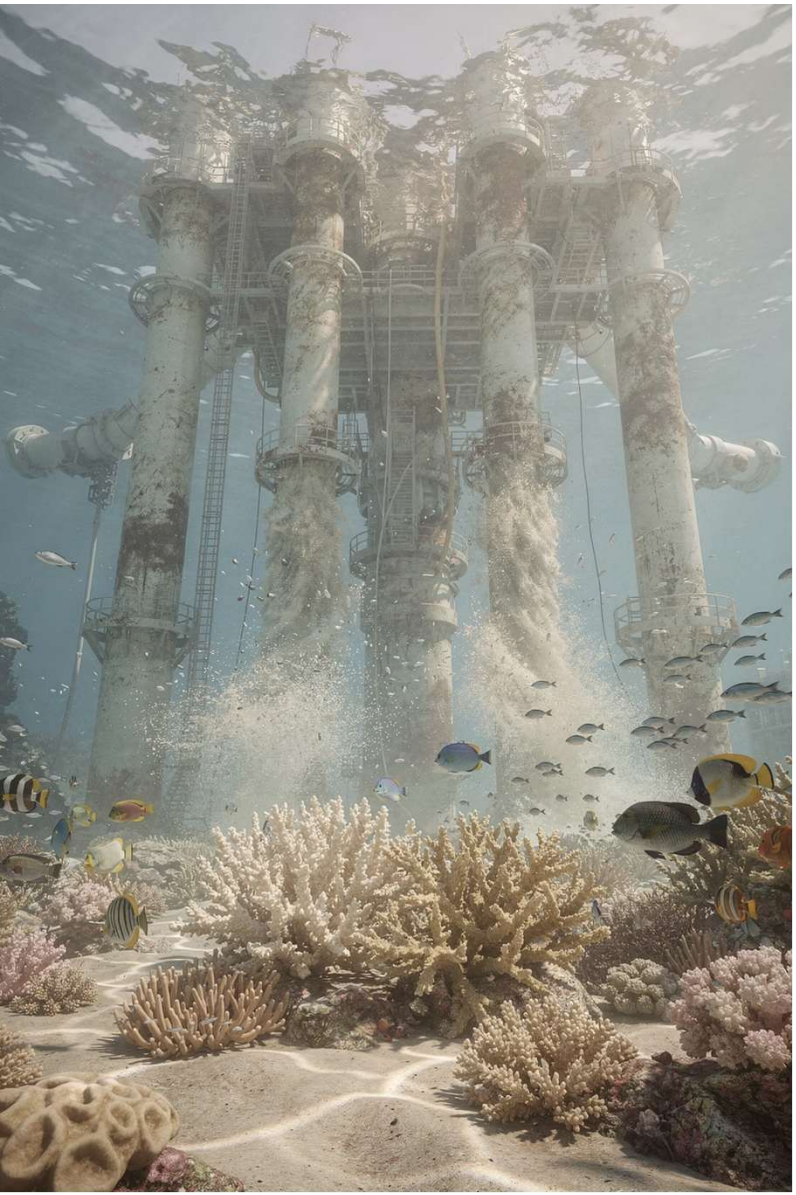
Ecosystem Enhancement

Deep cold water nutrient upwelling can naturally enhance local marine ecosystems and support healthier, more productive fisheries.



Ocean Cooling

Surface cooling effects may help mitigate coral bleaching events and potentially reduce hurricane intensity in warming seas.



Environmental Risks and Challenges

Temperature Alteration

Discharging large volumes of cold, nutrient-rich deep seawater can significantly alter local water temperature and chemical composition.

Marine Life Impact

Potential harm to marine organisms from intake pipe entrainment and sudden temperature shock in discharge zones.

Chemical Treatment

Use of chlorine compounds for biofouling control may affect water quality and sensitive marine species.

Disturbance Effects

Noise and light pollution near coastal installations can disrupt marine wildlife behaviour and nesting patterns.

Biofouling

Biofouling is the undesirable accumulation of microorganisms, algae, plants, and small animals on submerged surfaces, forming a complex biofilm.

It causes significant operational, economic, and environmental issues, such as increased hydrodynamic drag on ships, reduced heat exchanger efficiency, and the spread of invasive species.

Common on marine structures, pipes, and medical devices, it is managed through coatings, cleaners, and, in some cases, heat treatment.

Key mitigation strategies include using non-toxic coatings, surface modifications (silicone, nanotextured), chemical agents (chlorine), and, in industrial settings, biocides



Coral Bleaching

Coral bleaching occurs when corals, stressed by rising water temperatures, pollution, or extreme light, expel the colorful symbiotic algae (zooxanthellae) living in their tissues, turning them stark white.

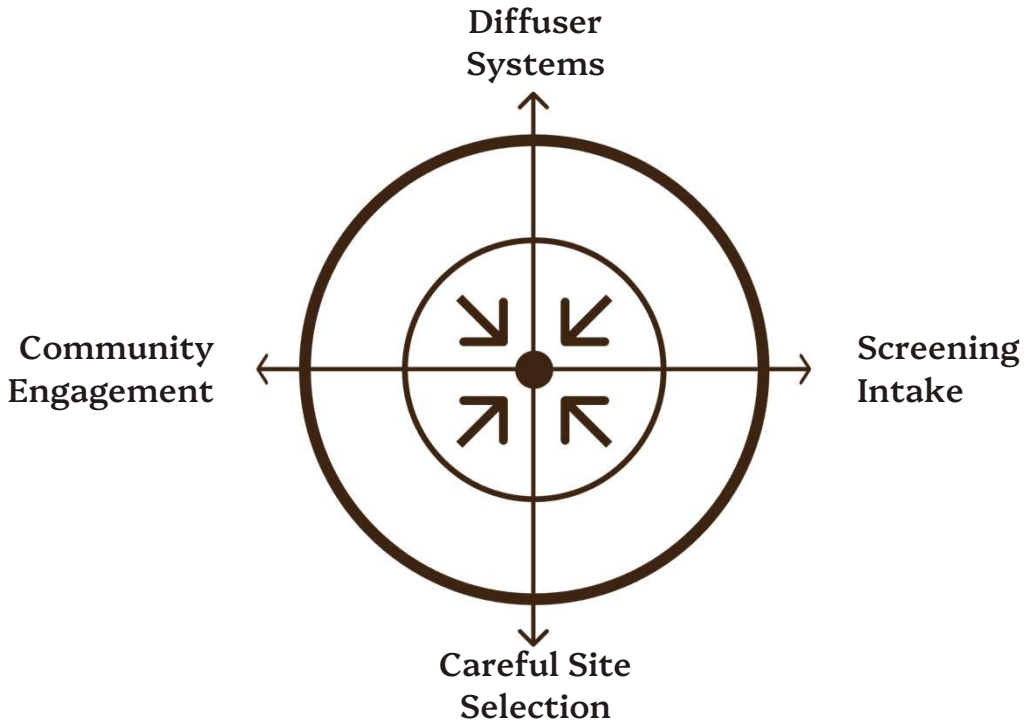
While not immediately dead, bleached corals are starving, highly vulnerable to disease, and likely to die if temperatures do not rapidly return to normal.

This phenomenon is driven primarily by climate change-induced heatwaves.

Corals rely on algae for food and color. When stressed, they expel these algae, leaving behind a translucent, white skeleton



Mitigation Strategies



Responsible OTEC development requires comprehensive mitigation strategies developed through scientific research and stakeholder collaboration.

- ***Diffusing discharged water at appropriate depths to minimise thermal shock and mixing impacts***
- ***Installing protective screening on intake pipes to significantly reduce marine organism entrainment***
- ***Conducting thorough site assessments to avoid sensitive habitats, spawning grounds, and protected areas***
- ***Engaging local communities and indigenous peoples in planning, monitoring, and benefit-sharing arrangements***

OTEC Environmental Protection Measures



CONCLUSION

OTEC's Role in a Sustainable Energy Future

Reliable Baseload Power

OTEC offers stable, renewable baseload electricity with valuable co-benefits including desalination, aquaculture opportunities, and coastal development.

Technological Progress

Ongoing technological advances and successful pilot projects worldwide are steadily paving the pathway towards commercial viability and widespread deployment.

Manageable Impacts

Environmental impacts are entirely manageable through proper engineering design, careful site selection, and robust regulatory frameworks.

Climate Solution

As climate change intensifies, OTEC could become a vital component of the global clean energy portfolio, especially for vulnerable tropical island nations seeking energy independence.